



<https://projects.mtconnect.org/account/register>

- Be part of a growing community.
- Provide valuable insight of your use cases for inclusion in the standard.
- Contribute to its development.

Resources

<https://www.mtconnect.org/>

<https://www.mtcup.org/>

<https://projects.mtconnect.org/>

<https://github.com/mtconnect>

<https://www.youtube.com/user/MTConnect>



MTCONNECT
VERSION 2.0
POWERED BY  **AMT**

Session 2

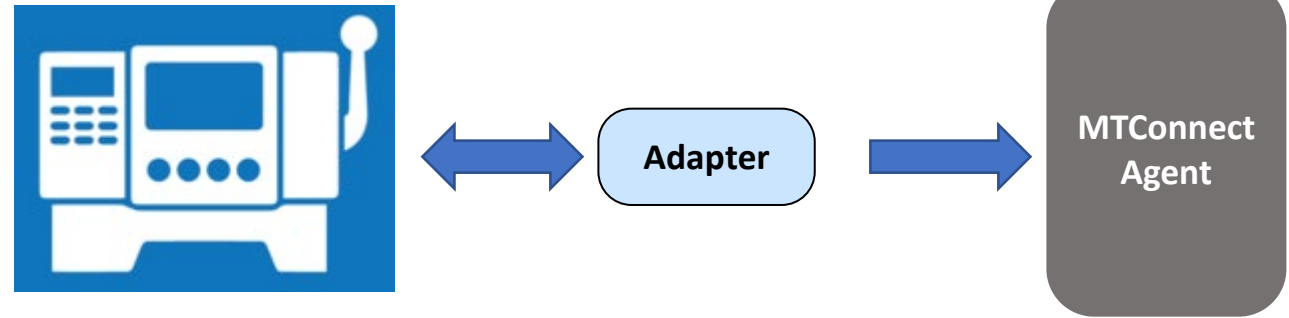
Adapters and Agents

Adapters and Agents

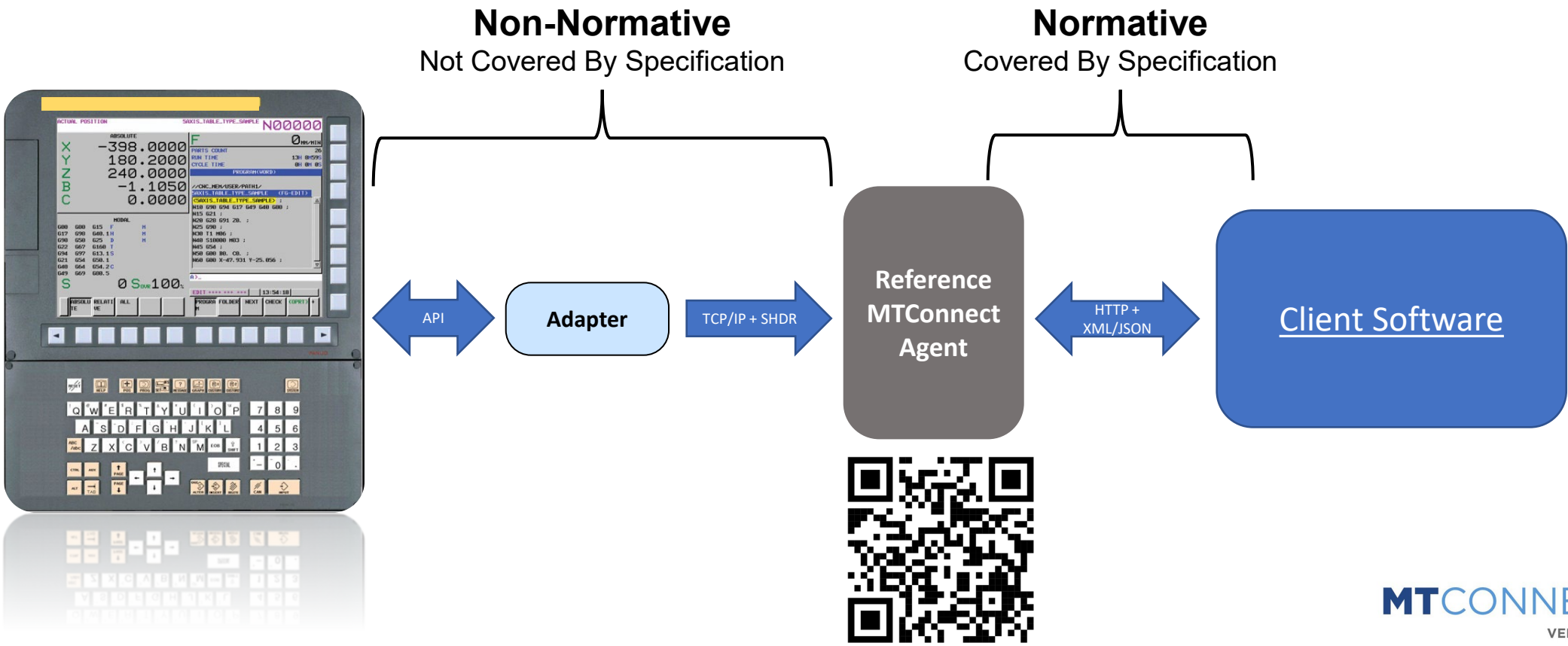
Roles & Relationships

What Are Adapters

1. Collect information from
 - Proprietary Interfaces
 - Electronic sources
2. Respond to Agent Connections
 - > * PING
 - < * PONG 10000
3. Push data to Agent
4. Recommendation: Use MTConnect vocabulary

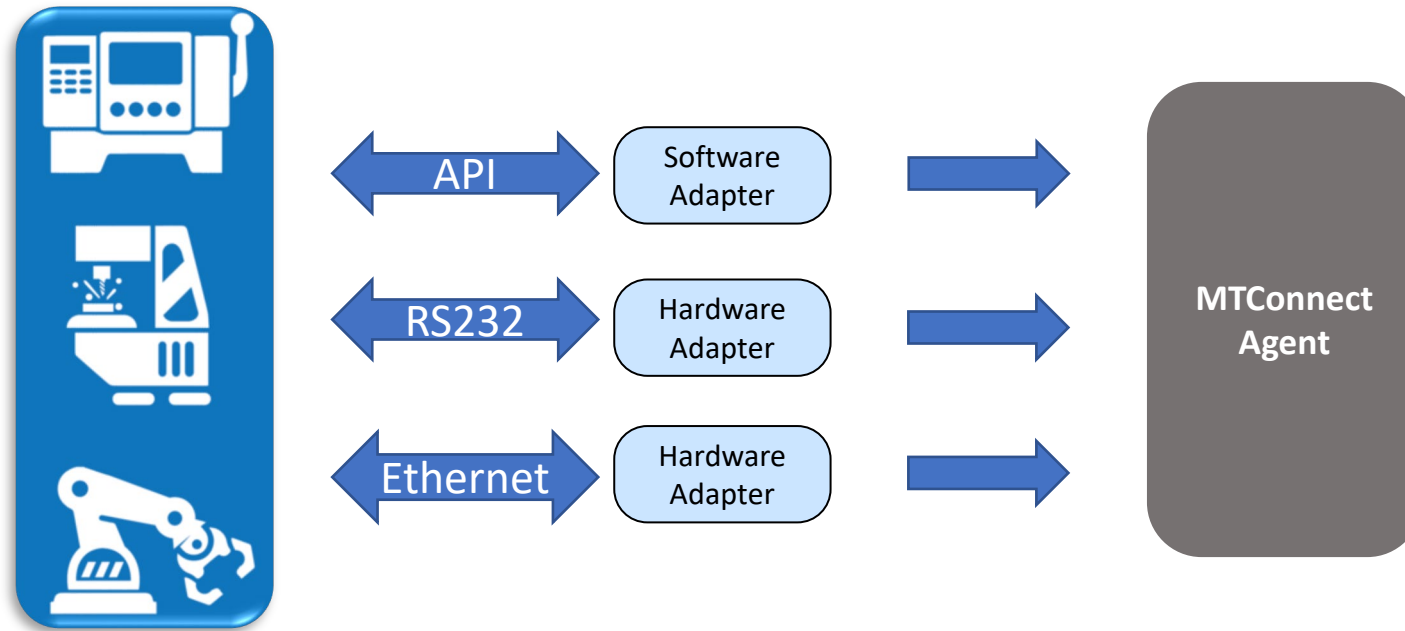


Why are Adapters not part of the MTConnect standard?



<https://github.com/mtconnect/cppagent>

How do Adapters work?



Adapter Output Format (SHDR v1.0)



Adapter

```
2014-01-16T04:13:38.704Z|Xtravel|NORMAL|||  
2014-01-16T04:13:38.704Z|Xabs|525.813|Xload|-6  
2014-01-16T04:13:38.704Z|Ytravel|NORMAL|||  
2014-01-16T04:13:38.704Z|Yabs|-283.839|Yload|-34  
2014-01-16T04:13:38.704Z|Ztravel|NORMAL|||  
2014-01-16T04:13:38.704Z|Zabs|657.125|Zload|-2|Srpm|0|Sload|4  
2014-01-16T04:13:38.704Z|Sload_cond|NORMAL|||
```

MTConnect
Agent



[https://github.com/mtconnect/cppagent/wiki/Simple_Hiearchial_Data_Representation_\(SHDR\)](https://github.com/mtconnect/cppagent/wiki/Simple_Hiearchial_Data_Representation_(SHDR))

MTCONNECT
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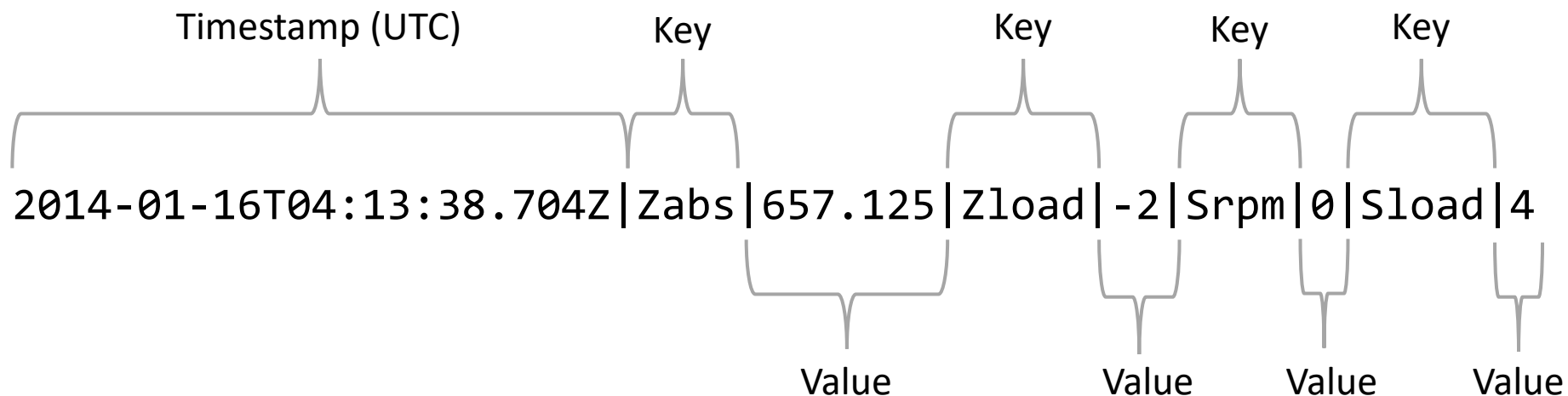
Adapter Output Format (SHDR 1.0)

Example of SAMPLE data

Timestamp (UTC) Key Key Key Key

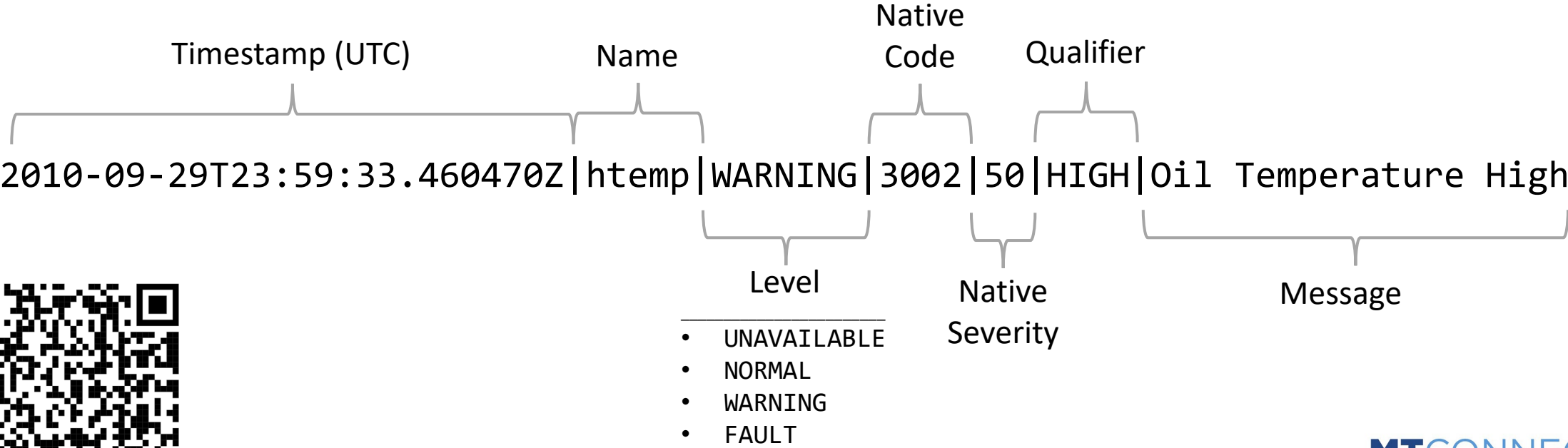
2014-01-16T04:13:38.704Z | Zabs | 657.125 | Zload | -2 | Srpm | 0 | Sload | 4

Value Value Value Value

A diagram illustrating the SHDR 1.0 data format. It shows a sequence of fields separated by vertical bars. The first field is a timestamp in UTC format. The subsequent fields are pairs of a key and a value. Brackets and labels above and below the fields identify their roles: 'Timestamp (UTC)' for the first field, 'Key' for the keys, and 'Value' for the values.

Adapter Output Format (SHDR v1.1)

Example of CONDITION data



Adapter Output Format (SHDR v1.1)

Example of MESSAGE data

Timestamp (UTC) Name Message

2010-09-29T23:59:33.460470Z | message | CHG_INSRT | Change Inserts

Native Code



[https://github.com/mtconnect/cppagent/wiki/Simple_Hiearchial_Data_Representation_\(SHDR\)#message](https://github.com/mtconnect/cppagent/wiki/Simple_Hiearchial_Data_Representation_(SHDR)#message)

Adapter Output Format (SHDR v1.2)

Example of TIME_SERIES data

Timestamp (UTC)	Name	Rate																	
2010-09-29T23:59:33.460470Z	temp	10	10	18.2	18.4	18.3	18.9	19.0	19.2	19.1	19.3	19.4	19.7						
		Count		Data															



Adapter Output Format (SHDR v1.2)

Example of ASSET data

Timestamp (UTC) Command Asset Type

2010-09-29T23:59:33.460470Z | @ASSET@ | KSSP300R.1 | CuttingTool | --multiline-0FED07ACED
<CuttingTool>
...
</CuttingTool>
--multiline-0FED07ACED

Asset ID Data...

Data Continued



Adapter Output Format (SHDR v1.2)

Example of ASSET data

Timestamp (UTC) Command

2010-09-29T23:59:33.460470Z | @REMOVE_ASSET@ | KSSP300R.1

Asset ID



Adapter Output Format (SHDR v1.2)

Example of ASSET data

Timestamp (UTC) Command

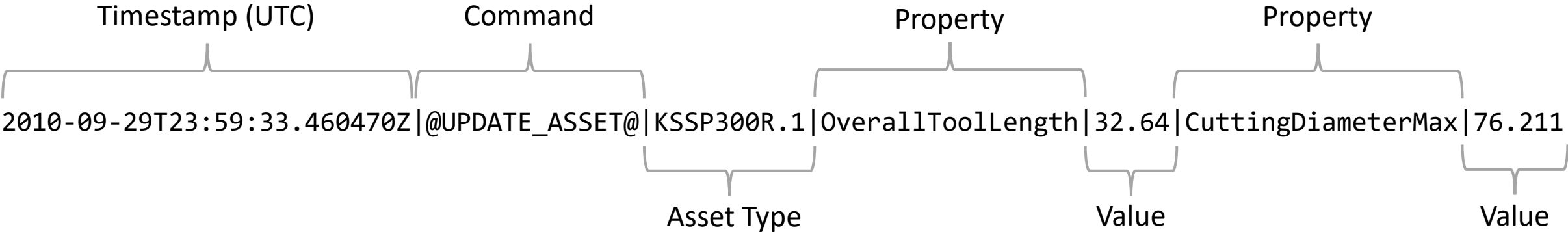
2010-09-29T23:59:33.460470Z | @REMOVE_ALL_ASSETS@ | CuttingTool

Asset Type



Adapter Output Format (SHDR v1.2)

Example of ASSET data



Adapter Output Format (SHDR v1.4)

Example of RESET_TRIGGER data

Timestamp (UTC)	Name	Value
2010-09-29T23:59:33.460470Z	pcount	0:DAY

Result



Adapter Output Format (SHDR v1.5)

Example of DATA_SET data

Timestamp (UTC) Name

2010-09-29T23:59:33.460470Z | vars | v1=10 v2=20 v3=30

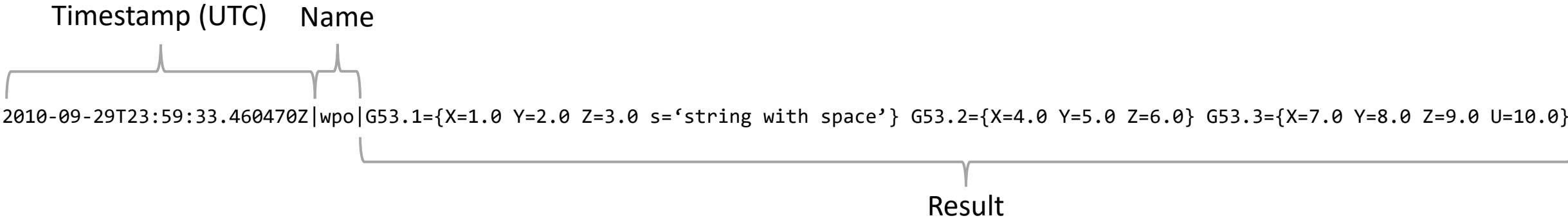
Result

- Add
- Remove
- Update



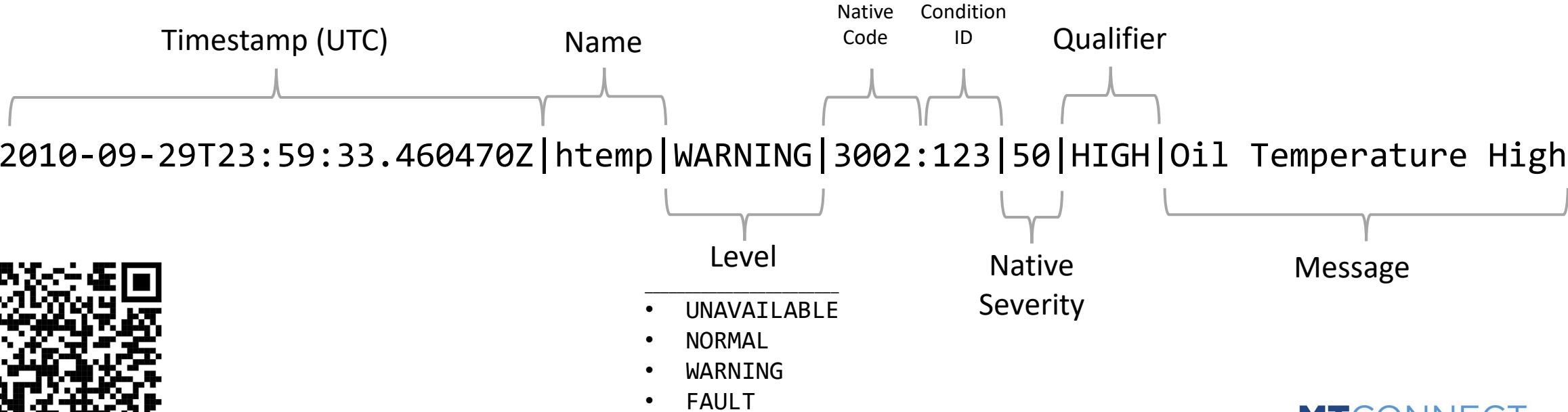
Adapter Output Format (SHDR v1.6)

Example of TABLE data



Adapter Output Format (SHDR v2.3)

Example of CONDITION data



[https://github.com/mtconnect/cppagent/wiki/Simple_Hiearchial_Data_Representation_\(SHDR\)#conditions-1](https://github.com/mtconnect/cppagent/wiki/Simple_Hiearchial_Data_Representation_(SHDR)#conditions-1)



Adapter > Agent Commands (Reference Agent)

Any data from Adapter Prefixed with and Asterisk (*) will not get parsed by the AGENT. It will be interpreted as a command.

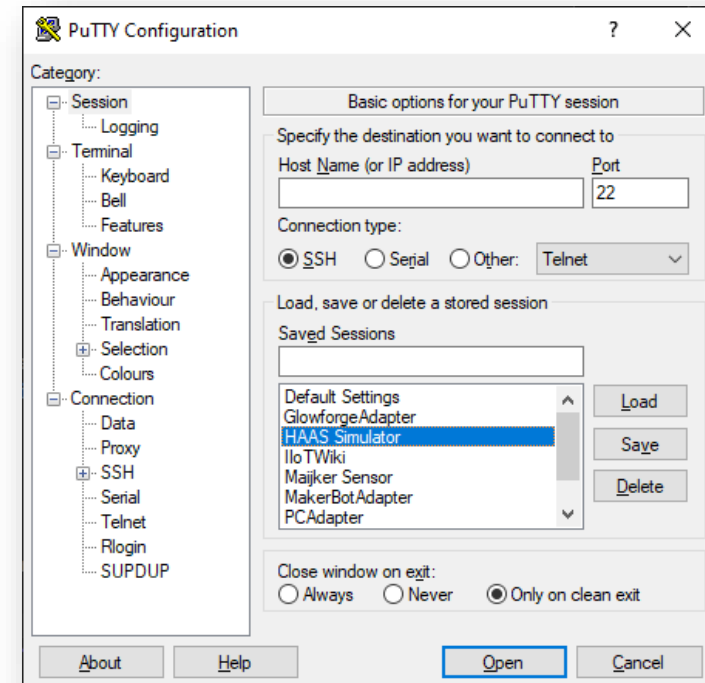
(Adapter > Agent)

```
> * adapterVersion: <version>
> * calibration: XXX
> * conversionRequired: <yes|no>
> * device: <uuid|name>
> * description: XXX
> * manufacturer: XXX
> * mtconnectVersion: <version>
> * nativeName: XXX
```

```
> * realTime: <yes|no>
> * serialNumber: XXX
> * shdrVersion: <version>
> * station: XXX
> * deviceModel: --multiline-
AAAAA\n<XML Fragment>\n--multiline-
AAAAA
> * PING
< * PONG 10000
```

Adapter Testing Tools

- Telnet
 - Windows: <https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/telnet>
 - Linux: <https://www.commandlinux.com/man-page/man1/telnet.1.html>
- PuTTY
 - Open-Source
 - Free Download: <https://www.putty.org/>
- IPAD/iPhone app (Mocha telnet)

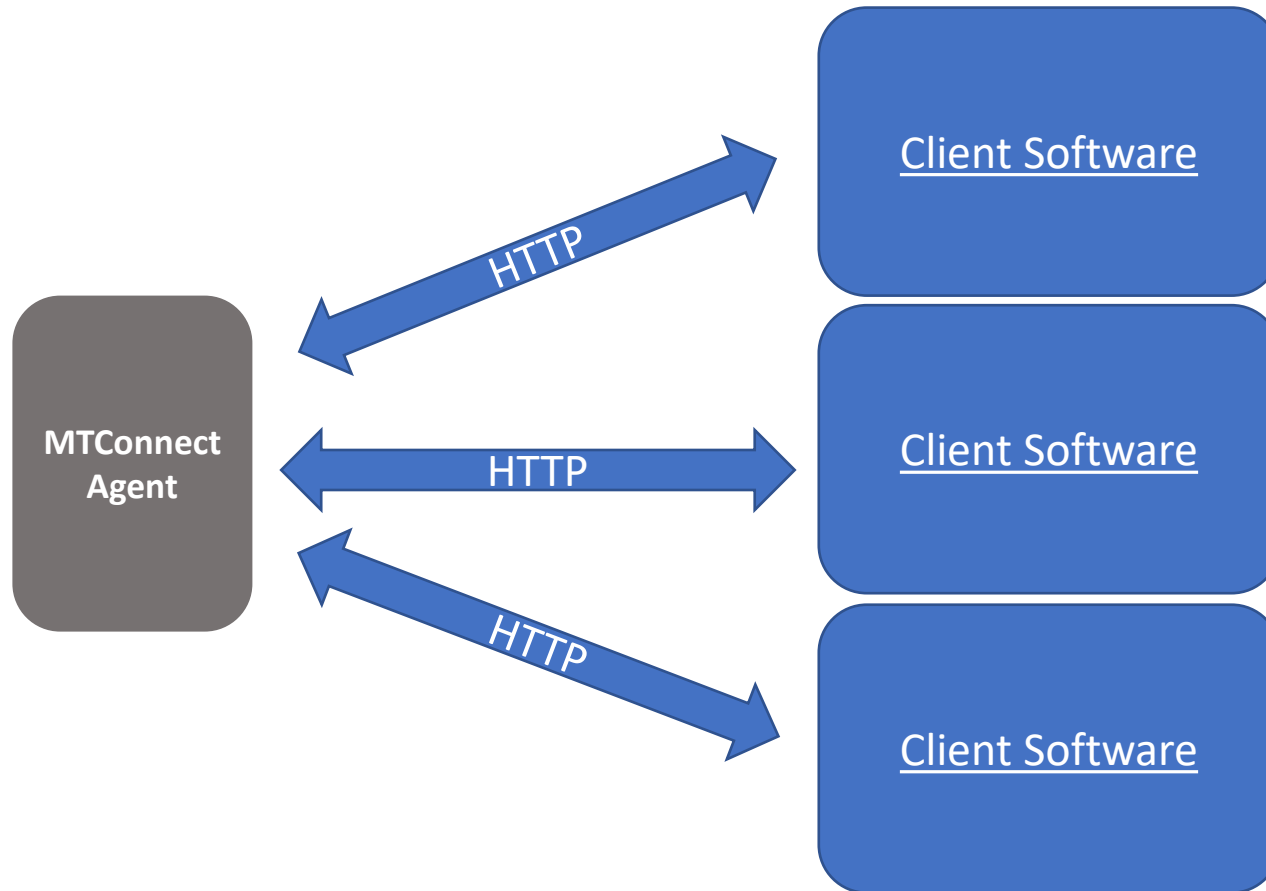


Lab - Connect to an Adapter

- Using PuTTY, open a connection to an existing adapter.
- What Mode is the controller in?
- What Execution Status is the Machine?
- Are there any Alarms?

What are Agents?

A process that implements the MTConnect® standard over HTTP, HTTPS, or MQTT.



HTTPS and MQTT with TLS are available in Agents v2.0 and higher

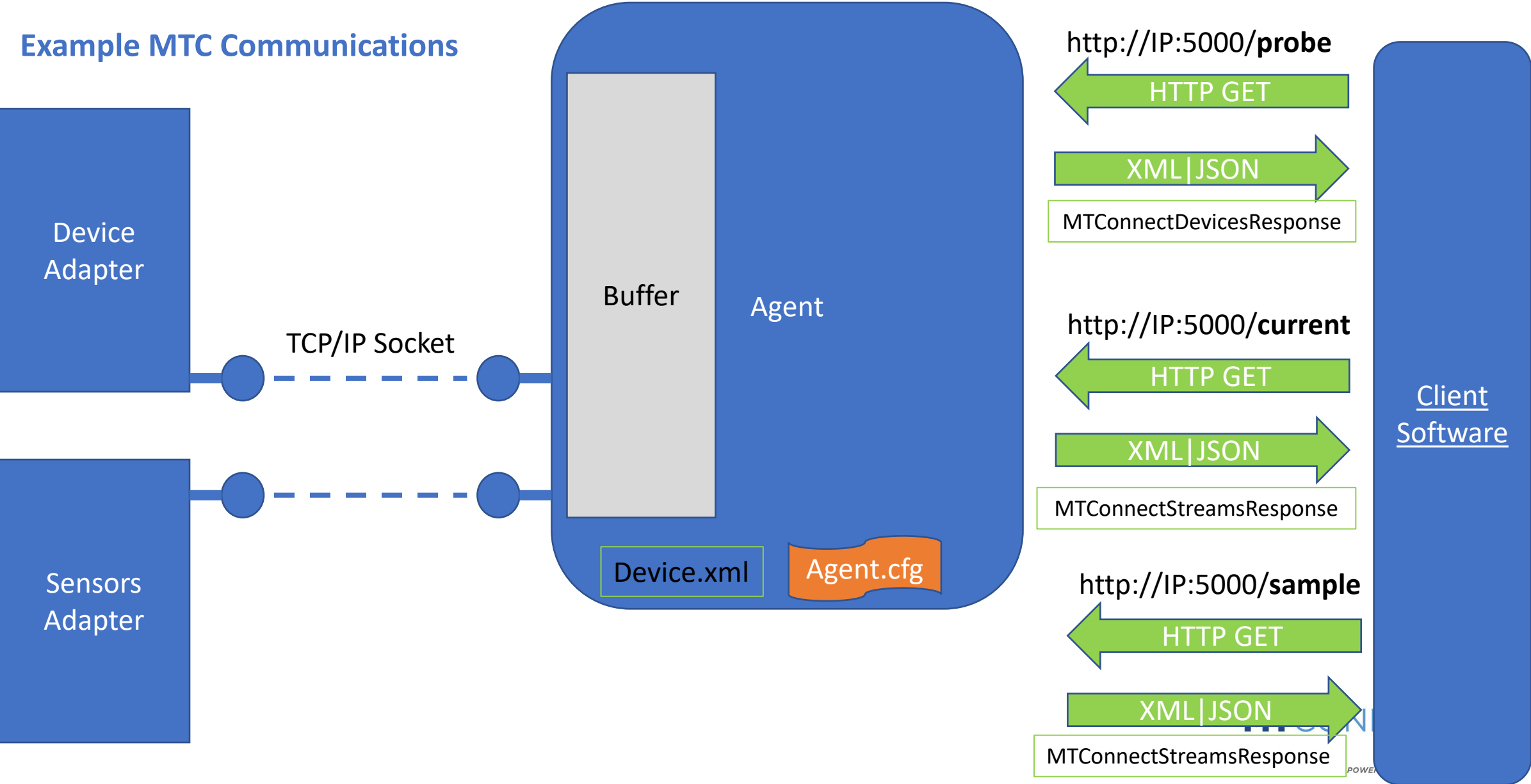
How do Agents Work

- Replies to HTTP GET.
- Responds with XML or JSON (based on Accept header).
- Stream response follows the MTConnect information model.



JSON formatted return documents are supported in Agents v1.5 and newer

Example MTC Communications



Agent Buffer

The information in MTConnect® can be thought of as a four-column table of data where the first column is a sequence number increasing by increments of one, the second column is the time, the third column is the data item it is associated with, and the fourth column is the value. The storage, internal representation, and implementation is not part of this standard. The implementer can choose to store as much or as little information as they want, as long as they can support the requirements of the standard. They can also decide if it is necessary to locally store the data.

Buffer size can be restricted in the AGENT CFG file.

Agent

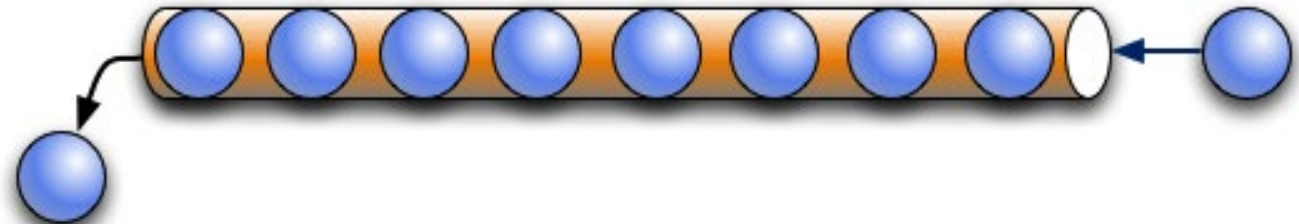
Seq	Time	Data Item	Value
101	2007-12-13T09:44:00.0221	Availability	UNAVAILABLE
102	2007-12-13T09:54:00.4412	Availability	AVAILABLE
103	2007-12-13T10:00:00.0002	Position Y	25
104	2007-12-13T10:00:00.0002	Position Z	1
105	2007-12-13T10:00:00.0002	Spindle Speed	0
106	2007-12-13T10:01:02.0012	Position X	11
107	2007-12-13T10:01:02.0012	Position Y	24
108	2007-12-13T10:01:02.0012	Position Z	1.1
109	2007-12-13T10:01:04.0012	Spindle Speed	1000
110	2007-12-13T10:01:04.5012	Position X	12
111	2007-12-13T10:01:04.5012	Position Y	23
112	2007-12-13T10:01:04.5012	Position Z	1.2
113	2007-12-13T10:01:05.5012	Position X	13
114	2007-12-13T10:01:05.5012	Position Y	22
115	2007-12-13T10:01:06.5012	Position X	14
116	2007-12-13T10:01:06.9012	Position Y	22
117	2007-12-13T10:01:07.0001	Position X	14
118	2007-12-13T10:01:07.0001	Position Z	1.3
119	2007-12-13T10:01:07.5001	Position X	15
120	2007-12-13T10:01:07.5001	Position Y	21
121	2007-12-13T10:01:07.5001	Position Z	1.4
122	2007-12-13T10:01:08.9012	Spindle Speed	0
123	2007-12-13T10:01:09.9012	Position X	10
124	2007-12-13T10:01:09.9012	Position Y	15
125	2007-12-13T10:01:09.9012	Position Z	0
126	2007-12-13T10:01:12.9012	Availability	UNAVAILABLE

Buffer Semantics (First In / First Out or “FIFO”)

Buffer filling up



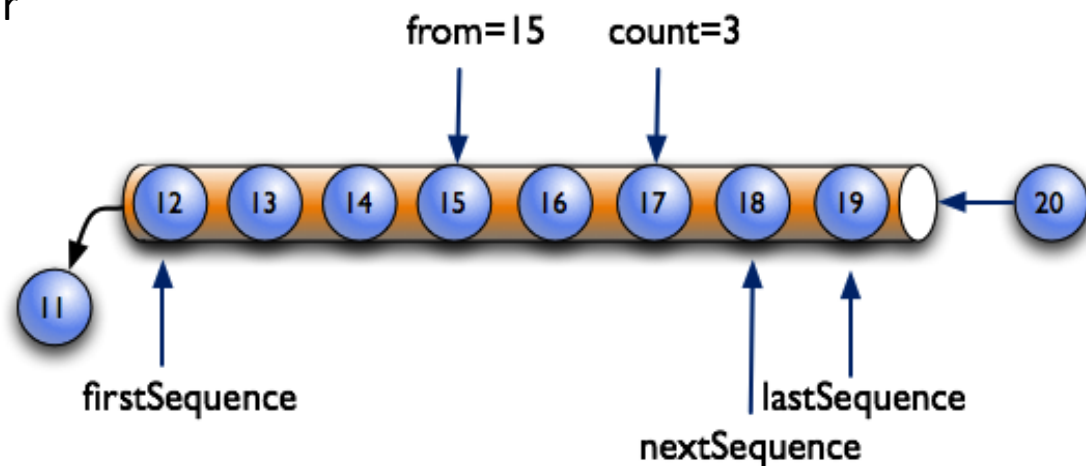
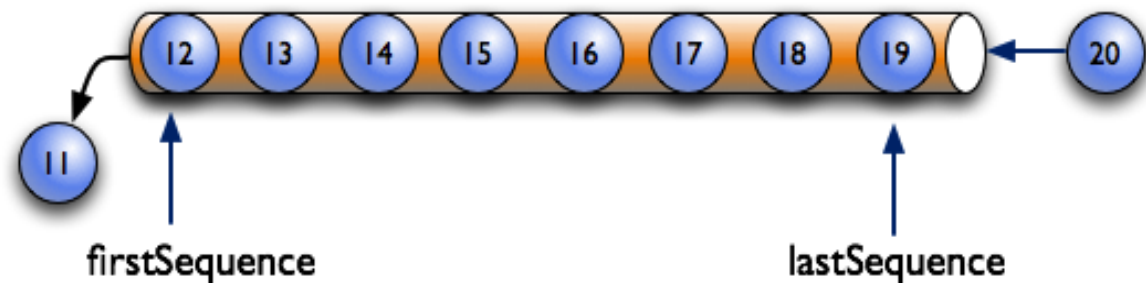
Buffer full and overflowing.



Buffer Semantics

How much Data is in the Buffer?

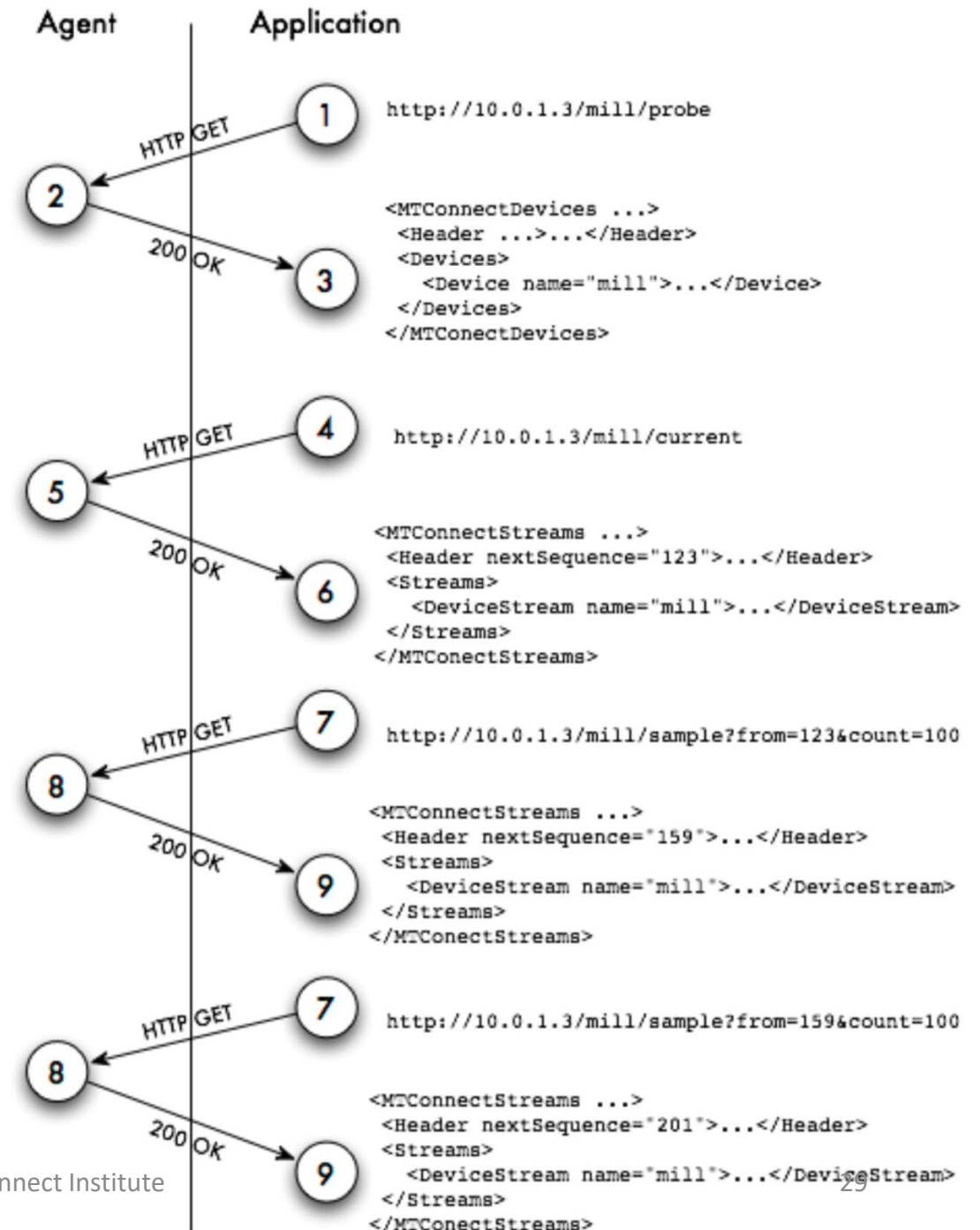
- **bufferSize**: The Max Number of Observations Agent buffer
- **firstSequence**: First available record in the agent buffer
- **lastSequence**: Last available record in the Agent buffer
- **nextSequence**: Next piece of data stored in the Agent buffer



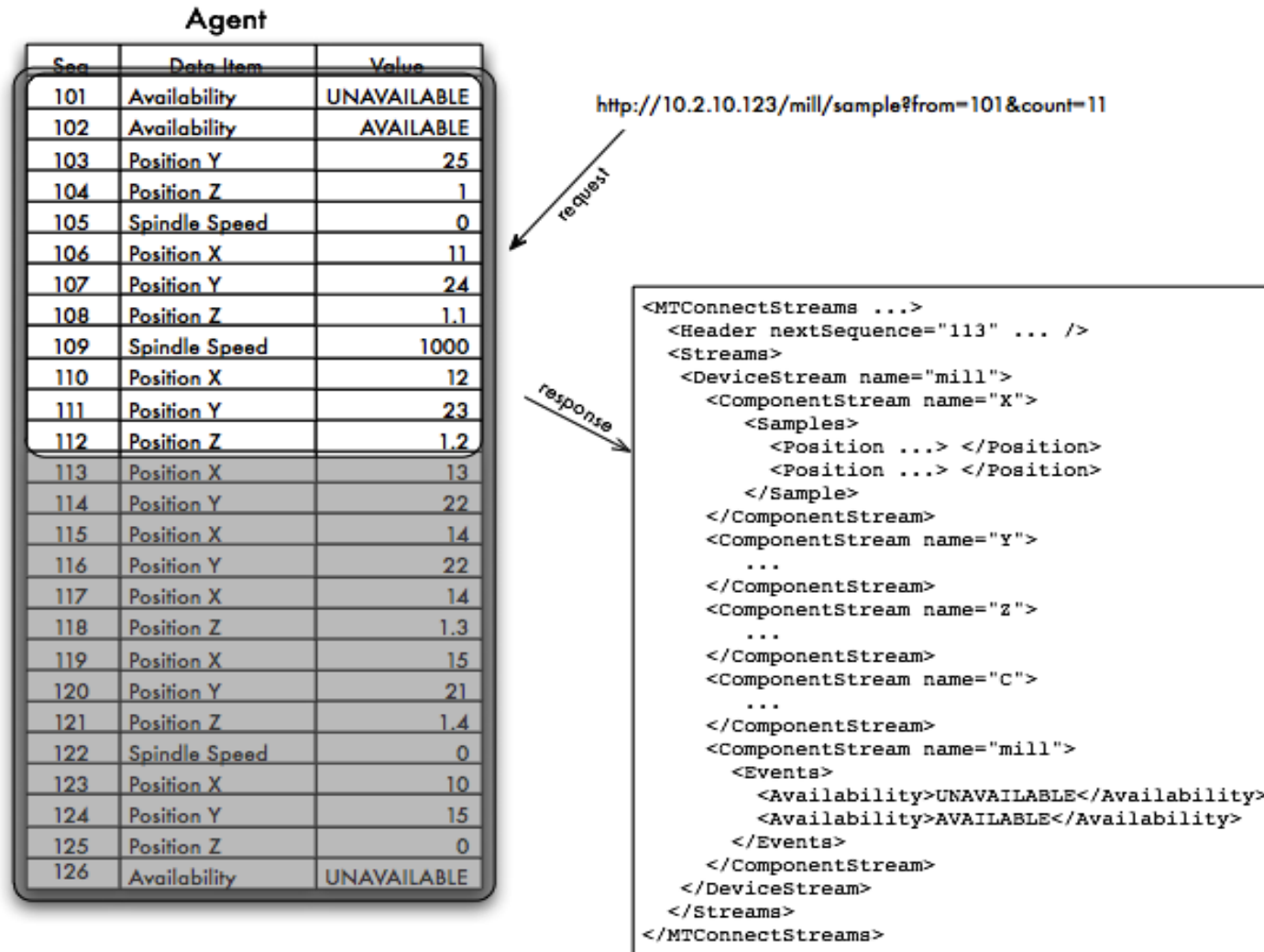
https://model.mtconnect.org/#Structure__EAID_2299B0B3_B209_47ae_B93A_BA694AD50051

Next Sequence Number

nextSequence tells you the (last location + 1) you retrieved data



Request without Filtering



Components

Logical or physical entity that provides a capability

Examples:

- Axis: X Y Z A B C J U V...
- Path
- ExtrusionUnit
- Door
- Tank

Data Item Types

Samples are point-in-time observations from a component reporting what the value is at that instant. The following are examples of Samples: Axis Position Data, Axis Loads, etc.

Events are state changes to a limited set of values or represent a message. It is assumed that an event remains in a state until the next occurrence of the event; it cannot have any intermediate values between the reported values. The following are examples of Events: Block, Execution, ControllerMode, Message etc.

A **Condition** communicates the device's health and ability to function (Categorized Alarms). It can be one of the following: UNAVAILABLE, NORMAL, WARNING, or FAULT, and multiple active conditions can exist at one time; whereas a sample or event can only have a single value at one point in time.

MTConnectDevicesResponse

(aka Devices.xml, probe response, or device configuration)

Live Demo - Review Devices.xml File

HTTP GET Requests, Agent software must support

MTConnect® *Agent* **MUST** support five types of requests:

probe – retrieves the components, data items, and configuration of the device. Returns a MTConnectDevices document.

current – retrieves a snapshot of the data item's most recent values or the state of the device at a point in time. Returns an MTConnectStreams document.

sample – retrieves the Samples, Events, and Condition in timeseries. Returns an MTConnectStreams document.

assets – requests the most recent state of an asset known to this device. Returns an MTConnectAssets document.

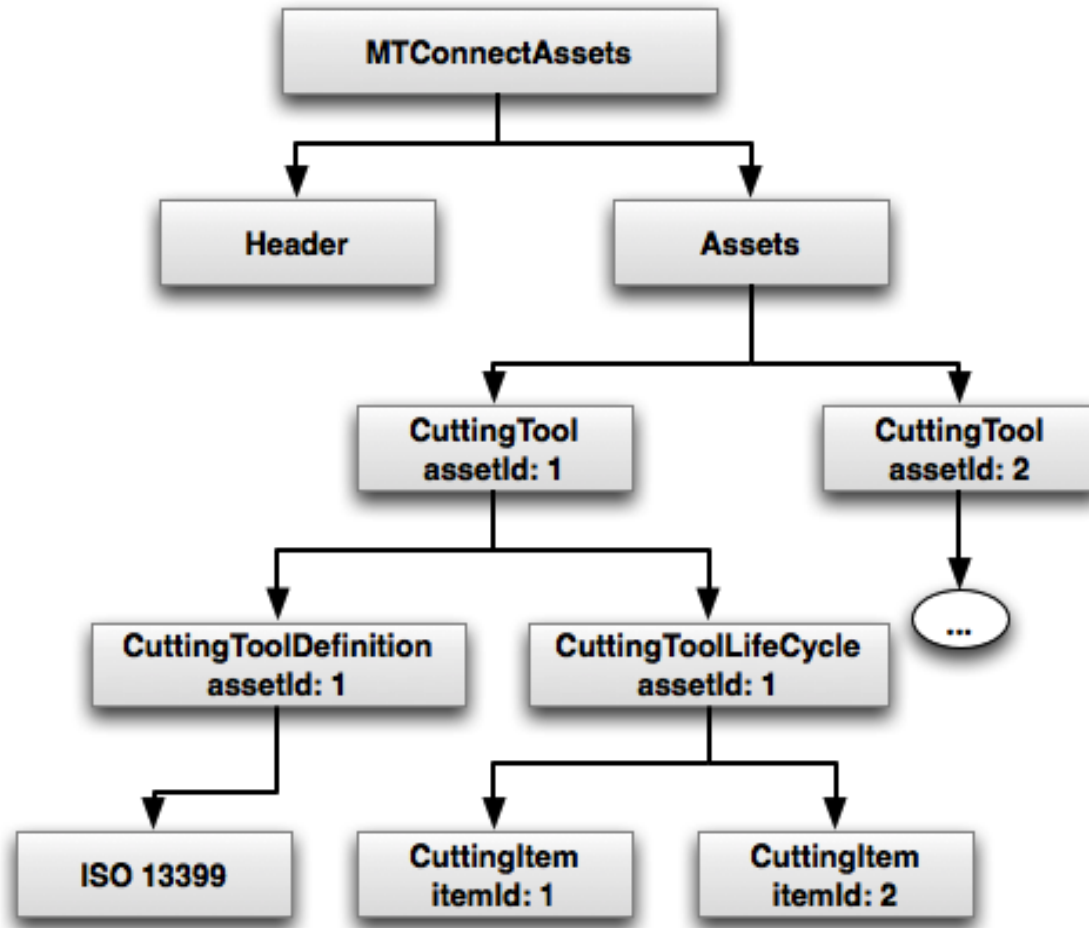
errors – must be returned by an agent when it encounters an error while interpreting a request for information from a client software or when an agent experiences an error while publishing the response to a request.

HTTP GET Requests, Agent software should support

If the MTConnect® *Agent* supports active validation of the device information model and pushed data, then this type of request **MUST** be supported:

exceptions – retrieves the list of validation exceptions that occurred in the respective response (Devices, Streams, or Assets).

Assets Collection (Separate from DataItems)



More recent Assets

- Files: lists of files, their mime type, URI, and other details.
- QIFDocument: Metrology data (on machine probing)

CuttingToolLifeCycle (example, from /assets)

```
<Assets>
<CuttingTool assetId="M8010W4194N1.110" deviceId="5fd88408-7811-3c6b-5400-11f4026b6890" serialNumber="0" timestamp="2022-05-05T10:34:34.183723Z"
toolId="14080012">
<CuttingToolLifeCycle>
  <CutterStatus>
    <Status>USED</Status>
  </CutterStatus>
  <Location negativeOverlap="0" positiveOverlap="0" type="POT">110</Location>
  <CuttingItems count="1">
    <CuttingItem indices="1">
      <Description>FACE MILL</Description>
      <ItemLife countDirection="UP" initial="0" limit="0" type="MINUTES" warning="80">1417.017</ItemLife>
      <ItemLife countDirection="UP" initial="0" limit="0" type="PART_COUNT" warning="80">151</ItemLife>
      <ProgramToolGroup>14080012</ProgramToolGroup>
      <Measurements>
        <CuttingDiameter nominal="80">80.158</CuttingDiameter>
        <FunctionalLength>381.144</FunctionalLength>
      </Measurements>
      <CutterStatus>
        <Status>USED</Status>
        <Status>AVAILABLE</Status>
        <Status>ALLOCATED</Status>
      </CutterStatus>
    </CuttingItem>
  </CuttingItems>
</CuttingToolLifeCycle>
</CuttingTool>
</Assets>
```

Files (example from /assets)

<Files> is an asset that lists various types of files on the device. In the case of CNC machines, you can expect part programs to be listed.

```
<Files>
<File applicationCategory="PROCESS" applicationType="PRODUCTION_PROGRAM" mediaType="text" name="92.EIA" size="18" versionId="0" state="PRODUCTION">
  <FileLocation href="file:///C:/ymw/M8Y/data/MC_Machine Programs/92.EIA"/>
  <CreationTime>2021-02-11T23:14:00.0000Z</CreationTime>
  <Destinations>
    <Destination>localhost</Destination>
  </Destinations>
</File>
<File applicationCategory="PROCESS" applicationType="PRODUCTION_PROGRAM" mediaType="text" name="93.EIA" size="63" versionId="0" state="PRODUCTION">
  <FileLocation href="file:///C:/ymw/M8Y/data/MC_Machine Programs/93.EIA"/>
  <CreationTime>2021-02-11T23:14:00.0000Z</CreationTime>
  <Destinations>
    <Destination>localhost</Destination>
  </Destinations>
</File>
<File applicationCategory="PROCESS" applicationType="PRODUCTION_PROGRAM" mediaType="text" name="95.EIA" size="145" versionId="0" state="PRODUCTION">
  <FileLocation href="file:///C:/ymw/M8Y/data/MC_Machine Programs/95.EIA"/>
  <CreationTime>2021-02-11T23:14:00.0000Z</CreationTime>
  <Destinations>
    <Destination>localhost</Destination>
  </Destinations>
</File>
</Files>
```

Q.I.F. Standard

Quality Information Framework (Q.I.F.) is a third part ISO metrology standard that is transmitted through the Agent Asset Buffer for things like on machine (probing) inspection results.

- Renishaw
- Mitutoyo
- Hexagon
- ZEISS

<https://qifstandards.org/>

Configuring Stand Alone Agents

<https://github.com/mtconnect/cppagent/releases>

Files that make up the AGENT

AGENT.EXE = Main Executable

AGENT.CFG = AGENT configuration file. (Read only at startup v1.2, file change event restart V1.3)

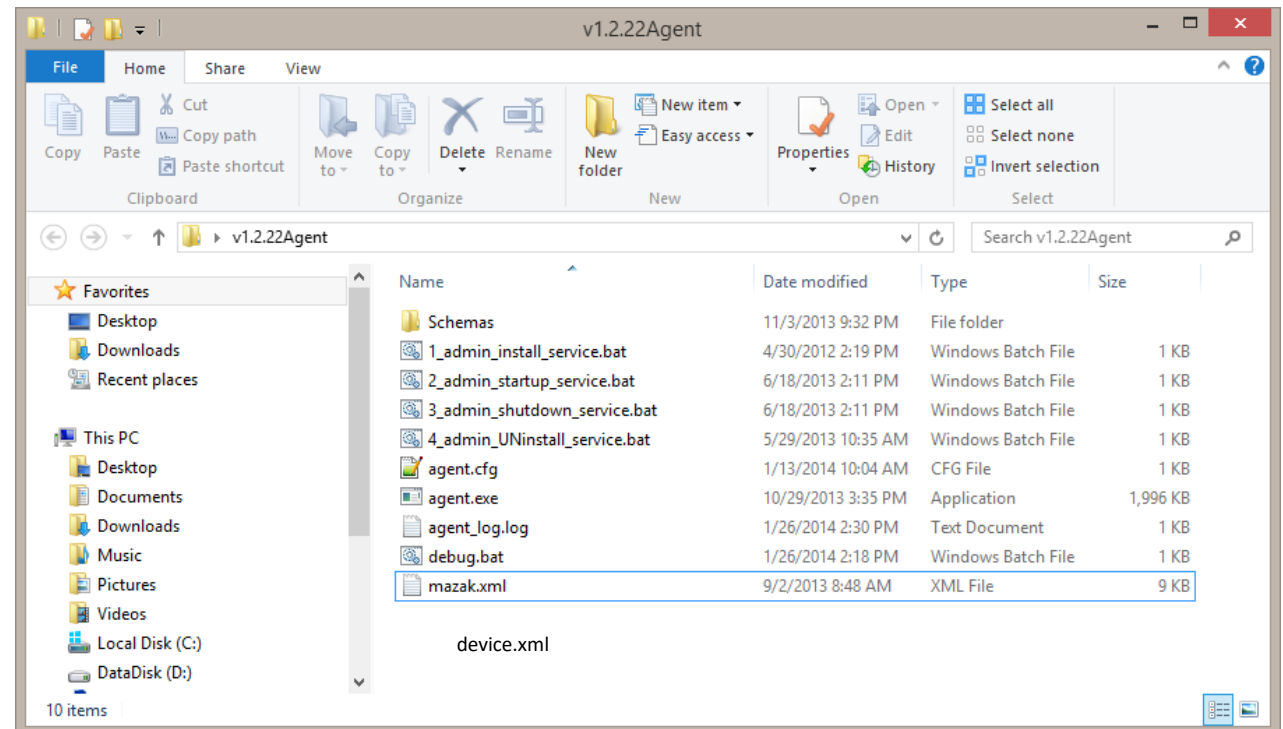
DEVICE.XML = Machine Configuration File (Read only at startup v1.2, file change event restart V1.3)

Agent_log.log = Error and info Logging file.

MonitorConfigFiles = true

Agent watches for file changing events and does a soft restart of the agent. Clears the buffer and increments the InstanceID.

Device info can now be updated without soft restart.



Desktop App vs. Windows Service

MTConnect Agent can be run as a Windows Desktop application or a Windows Service app. Agent is also available on LINUX and Mac OS.

Desktop Applications: Run under the security principle of the current Windows user that is logged on and does not run when no one is logged into the PC.

Windows Service Applications: Run under the Local SYSTEM account (or an assigned Domain User Account) and starts automatically when WINDOWS starts. It does not require anyone to be logged into the PC.

AGENT RUN Mode

RUN Starts the Agent in Run Mode. Runs as console Application.
All Logging is output to the log file.

```
C:\>cd C:\MTConnect\Agents\<Machine>\bin  
C:\MTConnect\Agents\<Machine>\bin>Agent.exe run
```

Debug Mode

DEBUG Starts the Agent in Debug Mode. Runs as a console Application.
All Logging is output to the console.

```
C:\>cd C:\MTConnect\Agents\<Machine>\bin  
C:\MTConnect\Agents\<Machine>\bin>Agent.exe debug
```

Installing as a “Windows Service” app

```
C:\>cd C:\MTConnect\Agents\<Machine>\bin  
C:\MTConnect\Agents\<Machine>\bin>Agent.exe install <agent.cfg>
```

❖ Allows user to override default settings

Using SC to install

SC is a command line program In WINDOWS used for communicating with the **Service Control Manager** and services.

```
sc create <ServiceName> obj=LocalSystem DisplayName=<Display Name> start=auto  
binpath=C:\MTConnect\Agents\<Machine>\bin\agent.exe
```

```
sc delete <ServiceName>
```

Agent Configuration File

```
Devices = Devices.xml
ServiceName = MTConnectAgentHAAS_5000
Port = 5000
BufferSize = 14
AllowPut = false
AllowPutFrom = localhost, 192.168.0.10
IgnoreTimestamps = true
PreserveUUID = false
ConversionRequired = false
MonitorConfigFiles = true
SchemaVersion = 2.7
Pretty = true
```

```
Adapters {
  18E {
    FilterDuplicates = yes
    Host = localhost
    Port = 7878
  }
}
```

Setting Parameters

Live Demo: Configuring AGENTS

Agent Testing Tools

- Internet Browser (Chrome, Edge, Firefox, Internet Explorer, Netscape Navigator, etc.)
- Postman
- Microsoft Excel
- Notepad++

Lab - Getting Data From Agents

Let's try these.

HTTP GET	Response Type	About
http://<ipaddress>:<port>	MTConnectDevicesResponse	MTConnect information model
http://<ipaddress>:<port>	MTConnectDevicesResponse	MTConnect information model
http://<ipaddress>:<port>/current	MTConnectStreamsResponse	Return last data for ALL tags
http://<ipaddress>:<port>/sample	MTConnectStreamsResponse	Return oldest 100 events in the Agent Buffer
http://<ipaddress>:<port>/assets	MTConnectAssetsResponse	Returns all assets from assets buffer

Instance ID

A unique number indicating the invocation of the *Agent*. This is used to differentiate between separate instances of the *Agent*.

InstanceID changes every time the Agent starts. This lets the Client know that something may have changed, or a failure may have occurred. (ID = Server Timestamp that agent started)

Client Apps should watch this value and PROBE the Agent to detect any changes.

**Buffer sequence numbers are reset when the InstanceID changes.*

Lab - cont.

`http://[ipaddress]:5000/current`

- Identify the Device name
- Find Availability status?
- What version of MTConnect Standard is being supported?
- Find the InstanceID.
- Find the Spindle Load.
- Find Estop Status.
- Find the Feedrate.

MTConnect Institute

- <https://www.mtconnect.org>
- <https://www.mtconnect.org/getting-started>
- <https://www.mtconnect.org/documentation>
- <https://model.mtconnect.org>
- <https://projects.mtconnect.org> (requires registration & login)
- The MTConnect Institute has a number of **Working Groups** dedicated to the Standard's development and promotion.
- E-Mail: bootcamp@projects.mtconnect.org

Q&A